

cp_3b

1.

Increment the number 🟡 20/20

● Solved

Easy

Write a program that reads a number N and checks if N is greater than 10.

Print the result of $N + 5$ if N is greater than 10. Otherwise, print the result of $N + 1$.

Input

The input will be a single line containing an integer representing N .

Output

The output should be a single line containing an integer. The result of $N + 5$ should be printed if N is greater than 10. Otherwise, the result of $N + 1$ should be printed.

Explanation

For example, if the given number is $N = 3$,

- 3 is **not** greater than 10.
- The result of $N + 1$ is 4 ($3 + 1 = 4$).

The output should be 4 as N is **not** greater than 10.

For example, if the given number is $N = 63$,

- 63 is greater than 10.
- The result of $N + 5$ is 68 ($63 + 5 = 68$).

The output should be 68 as N is greater than 10.

Sample Input 1

3

Sample Output 1

4

Sample Input 2

63

Sample Output 2

68

code:

```
N=int(input())
if(N>10):
    print(N+5)
else:
    print(N+1)
```

2.

Valid Triangle - 2 | 🟡 25/25

● Solved

Easy

Write a program that reads the three angles `A` , `B` , and `C` of a Triangle and checks if the sum of the three angles of the Triangle is equal to **180**.

Print the Triangle as given below if the sum of the three angles of the Triangle is equal to **180**. Otherwise, print **Not a Valid Triangle**.

```
*  
**  
***
```

Input

The first line of input contains an integer representing angle `A` .

The second line of input contains an integer representing angle `B` .

The third line of input contains an integer representing angle `C` .

Output

The output should be three lines containing a Triangle as shown in the sample output if the sum of `A` , `B` and `C` is equal to **180**.

Otherwise, the output should be a single line containing the string **Not a Valid Triangle**.

Explanation

For example, if the given three angles of the Triangle are `A = 60` , `B = 45` , and `C = 75` ,

- The sum of the three angles of the Triangle is equal to **180**. (`60 + 45 + 75 = 180`)
- The output should be the Triangle as given below,

```
*  
**  
***
```

To print the Triangle, print a star on the first line, two stars on the second line and three stars on the third line.

For example, if the given three angles of the Triangle are `A = 80` , `B = 90` and `C = 100` ,

- The sum of the three angles of the Triangle is **not** equal to **180**. (`80 + 90 + 100 = 270`)
- The output should be **Not a Valid Triangle**.

Sample Input 1

```
60  
45  
75
```

Sample Output 1

```
*  
**  
***
```

code:

```
A=int(input())  
B=int(input())  
C=int(input())  
if ((A+B+C)==180):  
    print("*")  
    print("**")  
    print("***")
```

```
else:
    print("Not a Valid Triangle")
```

3.

Greater than 100 | 🏆 25/25

● Solved

Easy

Write a program that reads three numbers `A` , `B` , and `C` , and checks if each number is greater than `100`.

Print **All are greater than 100** if each number is greater than `100`. Otherwise, print **Not all are greater than 100**.

Input

The first line of input contains an integer representing `A` .

The second line of input contains an integer representing `B` .

The third line of input contains an integer representing `C` .

Output

The output should be a single line containing a string. **All are greater than 100** should be printed if each number is greater than `100`. Otherwise, **Not all are greater than 100** should be printed.

Explanation

For example, if the given numbers are `A = 110` , `B = 256` , and `C = 6350` ,

✓ `A` is greater than `100`. (`110` is greater than `100`)

✓ `B` is greater than `100`. (`256` is greater than `100`)

✓ `C` is greater than `100`. (`6350` is greater than `100`)

The output should be **All are greater than 100** as each number is greater than `100`.

Sample Input 1

```
110
256
6350
```

Sample Output 1

```
All are greater than 100
```

Sample Input 2

```
120
50
1
```

Sample Output 2

```
Not all are greater than 100
```

code:

```
A=int(input())
B=int(input())
C=int(input())
if((A>100) and (B>100) and (C>100)):
    print("All are greater than 100")
```

```
else:
    print("Not all are greater than 100")
```

4.

Floor | 🟡 25/25

● Solved

Easy

Write a program that reads a Room Number and checks if the **Number** in the given Room Number is less than **30**.

The Room Numbers are in the format of **R1** , **R35** , etc.

Print **Ground Floor** if the **Number** is less than **30**. Otherwise, print **Not Ground Floor**.

Note

In the Room Number,

- The first character contains the character "R".
- The remaining characters contain a **Number**.

Example: **R1** , **R35** , etc.

- If the Room Number is **R35** , the **Number** in Room Number is **35**.

Input

The input will be a single line containing a string.

Output

The output should be a single line containing a string. **Ground Floor** should be printed if the **Number** in Room Number is less than **30**. Otherwise, **Not Ground Floor** should be printed.

Explanation

For example, if the given Room Number is **R27**,

R	2	7
0	1	2

- The first character of the Room Number is **R**.
- The remaining characters of the Room Number is **27**. So, the **Number** in Room Number is **27**.
- **27** is less than **30**.

The output should be **Ground Floor** as the room number is less than **30**.

Sample Input 1

R27

Sample Output 1

Ground Floor

code:

```
n=input()
if(int(n[1:])<30):
    print("Ground Floor")
```

```
else:  
    print("Not Ground Floor")
```

5.

Difference of Two Numbers 🟡 25/25

● Solved

Easy

Write a program that reads three numbers A , B , and C , and checks if the difference between any two numbers ($A - B$, $B - C$ and $C - A$) is **always** less than 25. Print **Difference is less than 25** if the difference between any two numbers ($A - B$, $B - C$ and $C - A$) is **always** less than 25. Otherwise, print **Difference is not less than 25**.

Input

The first line of input contains an integer representing A .
The second line of input contains an integer representing B .
The third line of input contains an integer representing C .

Output

The output should be a single line containing a string. **Difference is less than 25** should be printed if the difference between any two numbers is **always** less than 25. Otherwise, **Difference is not less than 25** should be printed.

Explanation

For example, if the given numbers are $A = 65$, $B = 60$, and $C = 52$,

The difference between A , B ($A - B$) and the difference between B , C ($B - C$) and the difference between C , A ($C - A$) should be less than 25.

✓ The difference $A - B$ is less than 25. ($65 - 60 = 5$, 5 is less than 25)

✓ The difference $B - C$ is less than 25. ($60 - 52 = 8$, 8 is less than 25)

✓ The difference $C - A$ is less than 25. ($52 - 65 = -13$, -13 is less than 25)

The output should be **Difference is less than 25** as the difference between any two numbers ($A - B$, $B - C$ and $C - A$) is **always** less than 25.

Sample Input 1

```
65  
60  
52
```

Sample Output 1

```
Difference is less than 25
```

Sample Input 2

```
100  
70  
35
```

Sample Output 2

```
Difference is not less than 25
```

code:

```
A=int(input())  
B=int(input())  
C=int(input())  
if((A-B)<25 and (B-C)<25 and (C-A)<25):  
    print("Difference is less than 25")
```

```
else:
    print("Difference is not less than 25")
```

6.

Valid Triangle - 3 | 🟡 30/30

● Solved

Write a program that reads three sides `A` , `B` , and `C` of a triangle and checks if the sum of any two sides of the triangle is **always** greater than the third side.

Print **It's a Triangle** if the sum of any two sides of the triangle is **always** greater than the third side. Otherwise, print **It's not a Triangle**.

Input

The first line of input contains an integer representing side `A` .
The second line of input contains an integer representing side `B` .
The third line of input contains an integer representing side `C` .

Output

The output should be a single line containing a string. **It's a Triangle** should be printed if the sum of any two sides of the triangle is **always** greater than the third side. Otherwise, **It's not a Triangle** should be printed.

Explanation

For example, if the given sides of the triangle are `A = 4` , `B = 5` and `C = 3` ,

The sum of sides `A` , `B` should be greater than `C` and the sum of sides `B` , `C` should be greater than `A` and the sum of sides `C` , `A` should be greater than `B` .

✓ `A + B` is greater than `C` . (`4 + 5 = 9` , `9` is greater than `3`)

✓ `B + C` is greater than `A` . (`5 + 3 = 8` , `8` is greater than `4`)

✓ `C + A` is greater than `B` . (`3 + 4 = 7` , `7` is greater than `5`)

The output should be **It's a Triangle** as the sum of any two sides of the triangle is **always** greater than the third side.

Sample Input 1

```
4
5
3
```

Sample Output 1

```
It's a Triangle
```

Sample Input 2

```
6
3
2
```

Sample Output 2

```
It's not a Triangle
```

code:

```
A=int(input())
B=int(input())
C=int(input())
if((A+B)>C) and ((B+C)>A) and ((C+A)>B):
    print("It's a Triangle")
```

```
else:
    print("It's not a Triangle")
```

7.

Valid String | 🟡 25/25

● Solved

Easy

Write a program that reads a string `S` and checks if the length of `S` is between **2** and **7** or the first character of `S` is **not equal** to **"a"**.

Print **Valid String** if the length of `S` is between **2** and **7** or the first character of `S` is **not equal** to **"a"**. Otherwise, print **Not a Valid String**.

Input

The input will be a single line containing a string representing `S`.

Output

The output should be a single line containing a string. **Valid String** should be printed if the length of `S` is between **2** and **7** or the first character of `S` is **not equal** to **"a"**. Otherwise, **Not a Valid String** should be printed.

Explanation

For example, if the given string `S = "apple"`,

✓ The length of `S` is between **2** and **7**. (The length of **apple** is **5**. **5** is between **2** and **7**)

✗ The first character of `S` is **not equal** to **"a"**. (The first character of **apple** is **equal** to **a**)

The output should be **Valid String** as the length of `S` is between **2** and **7**.

Sample Input 1

```
apple
```

Sample Output 1

```
Valid String
```

Sample Input 2

```
atlantic
```

Sample Output 2

```
Not a Valid String
```

Sample Input 3

```
out
```

Sample Output 3

```
Valid String
```

code:

```
s=input()
if((len(s)>2 and len(s)<7)or s[0]!="a"):
    print("Valid String")
else:
    print("Not a Valid String")
```

8.

Between 40 and 140 | 🟡 25/25

● Solved

Easy

Write a program that reads two numbers `A` and `B` and checks if any of the given numbers is between **40** and **140**.

Print **Between 40 and 140: Yes** if any of the given numbers is between **40** and **140**. Otherwise, print **Between 40 and 140: No**.

Input

The first line of input contains an integer representing `A`.

The second line of input contains an integer representing `B`.

Output

The output should be a single line containing a string. **Between 40 and 140: Yes** should be printed if any of the given numbers is between **40** and **140**. Otherwise, **Between 40 and 140: No** should be printed.

Explanation

For example, if the given numbers are `A = 12` and `B = 100`,

✗ `A` is between **40** and **140**. (**12** is **not** between **40** and **140**)

✓ `B` is between **40** and **140**. (**100** is between **40** and **140**)

The output should be **Between 40 and 140: Yes** as `B` is between **40** and **140**.

Sample Input 1

```
12
100
```

Sample Output 1

```
Between 40 and 140: Yes
```

Sample Input 2

```
33
4
```

Sample Output 2

```
Between 40 and 140: No
```

code:

```
A=int(input())
B=int(input())
first=(A>40) and (A<140)
second=(B>40) and (B<140)
c=first or second
if(c):
    print("Between 40 and 140: Yes")
else:
    print("Between 40 and 140: No")
```

9.

Sum or Numbers | 🟡 30/30

Easy

Write a program that reads two numbers `A` and `B`, and checks if one of the below conditions is satisfied.

- One of `A` and `B` is less than **20**.
- The sum of `A` and `B` is between **30** and **50**.

Print the sum of `A` and `B` if one of the given conditions is satisfied. Otherwise, print `A` and `B` on each line.

Input

The first line of input contains an integer representing `A`.
The second line of input contains an integer representing `B`.

Output

If one of the given conditions is satisfied,
The output should be a single line containing an integer that is the sum of `A` and `B`.
Otherwise,
The first line of output should be an integer containing `A`.
The second line of output should be an integer containing `B`.

Explanation

For example, if the given numbers are `A = 45` and `B = 7`,
✓ One of `A` and `B` is less than **20**. (`7` is less than **20**)
✗ The sum of `A` and `B` is between **30** and **50**. (`45 + 7 = 52`, **52** is **not** between **30** and **50**)
The output should be **52** (sum of `A` and `B`) as one of the given conditions is satisfied.

For example, if the given numbers are `A = 30` and `B = 100`,
✗ One of `A` and `B` is less than **20**. (`30` and `100` are **not** less than **20**)
✗ The sum of `A` and `B` is between **30** and **50**. (`30 + 100 = 130`, **130** is **not** between **30** and **50**)
The output should be,

```
30
100
```

Sample Input 1

```
45
7
```

Sample Output 1

```
52
```

code:

```
a=int(input())
b=int(input())
diff=(a<20) or (b<20)
d2=(a+b)>30 and (a+b)<50
op=diff or d2
```

```

if(op):
    print(a+b)
else:
    print(a)
    print(b)

```

10.

Lucky Number | 🏆 35/35

Write a program that reads two numbers `A` and `B` and checks if one of the below conditions is satisfied.

- One of `A` and `B` is equal to `6`.
- The sum of `A` and `B` is equal to `6`.
- The difference between `A` and `B` is equal to `6`.

Print **Lucky** if one of the given conditions is satisfied. Otherwise, print **Not Lucky**.

Note

Both `A - B` and `B - A` is used to calculate the difference between `A` and `B`.

Input

The first line of input contains an integer representing `A`.
The second line of input contains an integer representing `B`.

Output

The output should be a single line containing a string. **Lucky** should be printed if one of the given conditions is satisfied. Otherwise, **Not Lucky** should be printed.

Explanation

For example, if the given numbers are `A = 4` and `B = 10`,

- ▶ ✗ One of `A` and `B` is equal to `6`.
- ▶ ✗ The sum of `A` and `B` is equal to `6`.
- ▶ ✓ The difference between `A` and `B` is equal to `6`.

The output should be **Lucky** as one of the given conditions is satisfied.

Sample Input 1

```

4
10

```

Sample Output 1

```

Lucky

```

Sample Input 2

```

3
2

```

Sample Output 2

```

Not Lucky

```

code:

```
a=int(input())
b=int(input())
one=(a==6)or (b==6)
sum_6=(a+b)==6
diff=(a-b)==6 or(b-a)==6
if(one or sum_6 or diff):
    print("Lucky")
else:
    print("Not Lucky")
```

12.

Greatest Among Three Numbers | 🟡 35/35

Write a program that reads three numbers `A` , `B` and `C` and prints the greatest among the three numbers.

Input

The first line of input contains an integer representing `A` .
The second line of input contains an integer representing `B` .
The third line of input contains an integer representing `C` .

Output

The output should be a single line containing an integer that is the greatest among the three numbers.

Explanation

For example, if the given numbers are `A = 2` , `B = 5` and `C = 7` ,

- The greatest among `A` and `B` is `B` . (5 is greater than 2)
- The greatest among `B` and `C` is `C` . (7 is greater than 5)

The output should be `7` as it is the greatest among the three numbers.

Sample Input 1

```
2
5
7
```

Sample Output 1

```
7
```

Sample Input 2

```
3
5
2
```

Sample Output 2

```
5
```

code:

```
a=int(input())
b=int(input())
c=int(input())
if((a>=b) and (a>=c)):
    print(a)
if((b>a) and (b>c)):
    print(b)
if((c>a) and (c>b)):
    print(c)
```

13.

Bonus Salary | 🟡 25/25

● Solved

A company decided to give a bonus of 5% to an employee if his/her years of service is more than five years.

Write a program that reads an employee's salary and years of service and decides whether the employee gets the bonus or not.

Input

The first line of input will contain the salary of an employee.

The second line of input will contain years of service.

Output

If the employee gets a bonus, print the net bonus amount.

If the employee doesn't get the bonus, print "No Bonus",.

Explanation

For example, if the employee's salary is 25000 and years of experience is 3. As the years of experience is less than 5, the output should be "No Bonus".

Similarly, if the employee's salary is 50000 and years of experience is 6. As the years of experience is more than 5, the employee is eligible for the bonus. By computing the 5% of his salary, the net bonus amount should be 2500.0

Sample Input 1

```
25000
3
```

Sample Output 1

```
No Bonus
```

Sample Input 2

```
50000
6
```

Sample Output 2

```
2500.0
```

code:

```
salary=int(input())
experience=int(input())
if experience >5:
    bonus=salary*0.05
    print(bonus)
else:
    print("No Bonus")
```

14.

Valid Triangle | 25/25

● Solved

Given three angles of a triangle, write a program to check whether the triangle is valid or not. A triangle is valid if the sum of its three angles is equal to 180 degrees.

Input

The first line of input will contain the first angle of the triangle.

The second line of input will contain the second angle of the triangle.

The third line of input will contain the third angle of the triangle.

Output

If the sum of the three angles is equal to 180, print "It's a Triangle". In all other cases, print "It's not a Triangle".

Explanation

For example, if the first angle is 80, the second angle is 90, and the third angle is 100. Therefore the sum of the three angles ($80 + 90 + 100$) is 270, which is not equal to 180. So the output should be "It's not a Triangle".

Similarly, if the first angle is 60, the second angle is 60, and the third angle is 60. Therefore the sum of the three angles ($60 + 60 + 60$) is 180, which is equal to 180. So the output should be "It's a Triangle".

Sample Input 1

```
80
90
100
```

Sample Output 1

```
It's not a Triangle
```

Sample Input 2

```
60
60
60
```

Sample Output 2

```
It's a Triangle
```

code:

```
a=int(input())
b=int(input())
c=int(input())
if((a+b+c)==180):
    print("It's a Triangle")
else:
    print("It's not a Triangle")
```